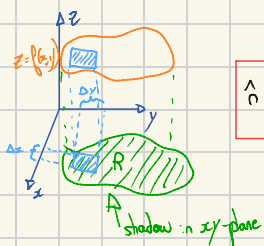


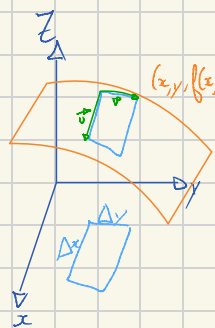
Lecture 28: Divergence Theorem

Flux of $\vec{F}$ through surface S is $\iint_S \vec{F} \cdot \hat{n} dS$

* If S is a graph of $z = f(x, y)$



$$\hat{n} dS = \pm \langle f_x, f_y, 1 \rangle dx dy$$



$$\pm \vec{u} \times \vec{v} = \Delta S \cdot \hat{n}$$

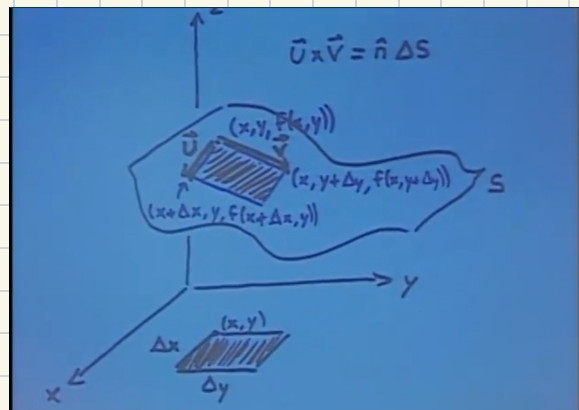
$$\vec{u} : \text{from } (x, y, f(x, y)) \text{ to } (x + \Delta x, y, f(x + \Delta x, y))$$

$$\vec{v} : \text{from } (x, y, f(x, y)) \text{ to } (x, y + \Delta y, f(x, y + \Delta y))$$

$$\vec{u} \approx \langle \Delta x, 0, f_x \cdot \Delta x \rangle = \langle 1, 0, f_x \rangle \cdot \Delta x$$

$$\vec{v} \approx \langle 0, \Delta y, f_y \cdot \Delta y \rangle = \langle 0, 1, f_y \rangle \cdot \Delta y$$

$$\hat{n} dS = \vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix} \Delta x \Delta y = \langle f_x, f_y, 1 \rangle \Delta x \Delta y \quad (\text{with } \hat{n} \text{ 'upwards'})$$

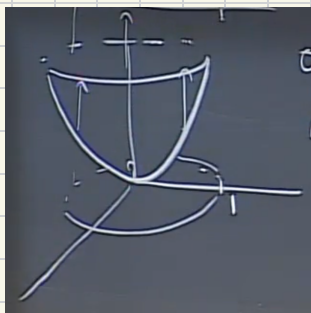


Example: $\vec{F} = z\hat{k}$ through portion of paraboloid $z = x^2 + y^2$ above unit disk.
with $\hat{n}$ pointing upwards (Flux should be > 0)
 $f = x^2 + y^2$

$$\iint_S \vec{F} \cdot \hat{n} dS = \iint_S \langle 0, 0, z \rangle \cdot \langle -2x, -2y, 1 \rangle dx dy$$

$$= \iint_S z dx dy$$

$$= \iint_S (x^2 + y^2) dx dy = \int_0^{2\pi} \int_0^1 r^2 \cdot r dr d\theta = \frac{\pi}{2}$$



More generally: given a parametric equation

$$S: \begin{cases} x = x(u, v) \\ y = y(u, v) \\ z = z(u, v) \end{cases} \quad (x, y, z) = \vec{r} = \vec{r}(u, v)$$

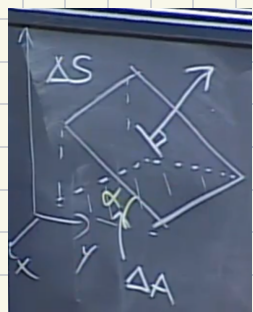
$$\hat{n} dS = \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du dv = \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) du dv$$



$$\text{Sides: } \frac{\partial \vec{r}}{\partial u} \Delta u \text{ and } \frac{\partial \vec{r}}{\partial v} \Delta v$$

* If we know a normal vector $\vec{N}$ (not necessarily a unit one)

Examples: * plane $ax + by + cz = d$ then $\vec{N} = \langle a, b, c \rangle$
* Surface given by eq. $g(x, y, z) = 0$ then $\vec{N} = \nabla g$



Surface element, $\Delta A = \Delta S \cos \alpha$

$$\cos \alpha = \frac{\vec{N} \cdot \hat{k}}{|\vec{N}| |\hat{k}|} = \frac{\vec{N} \cdot \hat{k}}{|\vec{N}|}$$

$$\Delta S = \frac{1}{\cos \alpha} \Delta A = \frac{|\vec{N}|}{\vec{N} \cdot \hat{k}} \Delta A$$

$$\hat{n} dS = \frac{|\vec{N}| \hat{n}}{\vec{N} \cdot \hat{k}} = \pm \frac{\vec{N}}{\vec{N} \cdot \hat{k}} \Delta A$$

$$\hat{n} dS = \pm \frac{\vec{N}}{\vec{N} \cdot \hat{k}} dx dy$$

Example: $S: z = f(x, y) = 0$

$$\vec{N} = \nabla g = \langle -f_x, -f_y, 1 \rangle$$

$$\frac{\vec{N}}{\vec{N} \cdot \hat{k}} dx dy = \langle -f_x, -f_y, 1 \rangle dx dy$$

Divergence Theorem: ("Gauss-Green" Theorem) (3D analogue of Green's Thm for flux)

If S is a closed surface enclosing a region D , oriented with $\hat{n}$ outwards and $\vec{F}$ defined and differentiable everywhere in D , then:

$$\iint_S \vec{F} \cdot \hat{n} dS = \iiint_D \text{div } \vec{F} dV \quad \text{where } \text{div}(\vec{P} + \vec{Q} + \vec{R}) = \vec{P}_x + \vec{Q}_y + \vec{R}_z$$

Example: last time: flux of $z\hat{k}$ through sphere radius a was $\frac{4}{3}\pi a^3$

$$\iint_S z\hat{k} \cdot \hat{n} dS = \iiint_D \underbrace{\text{div}(z\hat{k})}_{0+0+1=1} dV = \iiint_D dV = \text{volume}(D)$$